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INTEGRATED SUPPLY CHAIN OPTIMIZATION OF OPEN-PIT MINING HAULAGE SYSTEMS: AN AGENT-BASED SIMULATION OF TRUCK–SHOVEL OPERATIONS UNDER STOCHASTIC AVAILABILITY

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ABSTRACT. In today's large open-pit mines, the truck–shovel system sits at the heart of the upstream supply chain, directly feeding crushers, processors, stockpiles, and ultimately customers. Any delay or imbalance here quickly ripples downstream. To address this, we developed a high-fidelity, data-driven Agent-Based Simulation (ABS) framework that works hand-in-hand with simulation–optimization. The model treats trucks, shovels, and loading/dumping points as intelligent agents that react in real time to congestion, equipment readiness, changing mine geometry, and stochastic failures. By feeding the simulation with actual reliability data from Golgohar Iron Ore Mine No. 1, we captured real-world variability far more accurately than traditional deterministic or discrete-event approaches. The results are clear and actionable: the supply chain is mainly bottlenecked by truck availability. A coordinated strategy—raising truck availability by 30 % and shovel availability by 10 %—delivers a 31.4 % jump in total material moved, cuts average queue length by 13 %, and keeps the loading and hauling subsystems nicely balanced. Monte Carlo validation across 1,000 runs confirms these gains are statistically robust. In short, blending decentralized agent behavior, real operational data, and joint optimization gives mine managers a practical, reliable tool for strengthening the entire mining supply chain.

1. Introduction. From a supply chain viewpoint, the truck–shovel system is not merely an isolated operational unit but a pivotal upstream node that regulates material flow toward crushing, processing, stockpiling, and ultimately market delivery. Disruptions or inefficiencies in haulage propagate downstream, amplifying variability in processing schedules, inventory levels, and contractual fulfillment performance. Therefore, optimizing truck–shovel interactions must be understood as a supply chain coordination problem rather than a purely local dispatching challenge. By explicitly modeling decentralized equipment behavior, stochastic availability, and spatial constraints within an integrated simulation environment, this research bridges operational haulage optimization and mining supply chain performance, offering a systemic framework for aligning upstream extraction capacity with downstream processing and distribution requirements. Efficient haulage operations represent one of the most critical components of open-pit mining systems, where the truck–shovel network forms the backbone of material transportation and strongly influences overall extraction performance [1]. As global demand for iron ore increases and mining operations expand in scale and complexity, achieving higher production targets requires strategies that reduce operational delays, improve equipment utilization, and contain rising haulage costs [2]. In large open-pit mines such as Golgohar Iron Ore Mine No. 1, haulage operations account for nearly 50–60% of total operating expenditures [3], highlighting the strategic importance of optimizing fleet performance. However, haulage optimization remains a challenging task due to the highly dynamic nature of open-pit environments [4]. Continuous changes in mine geometry, shifting haul routes, variable loading dumping

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patterns, and frequent equipment failures introduce substantial operational uncertainty [5], [6]. Additional factors including queue formation, imbalances between truck and shovel capacity, and non-stationary cycle times further reduce system efficiency [7], [8]. Traditional optimization approaches often rely on static assumptions and fail to capture the decentralized, adaptive, and interactive behaviors observed in real fleet operations [9]. These challenges underscore the need for data driven, flexible, and behavior aware modeling frameworks capable of representing the complexity of modern haulage systems. Accordingly, this study addresses the following research questions:

i. How do stochastic availability patterns of trucks and shovels reshape haulage system bottlenecks?
ii. To what extent do joint improvements in loading and hauling availability outperform single-subsystem optimization strategies?

2. Literature review. Extensive research in mine optimization consistently identifies haulage planning as one of the most influential factors affecting production performance and operational cost [10], [11]. Conventional analytical and simulation-based methods such as linear programming and Discrete-Event Simulation (DES) have been widely applied to evaluate truck–shovel operations, queue dynamics, and loading hauling cycles [12], [13]. While DES provides valuable insights into event-driven processes, its ability to represent decentralized decision-making and emergent system behavior remains limited [14], [15].

To address these limitations, Agent-Based Modeling (ABM) has been introduced as a powerful alternative for modeling complex mining systems [16]. ABM enables each equipment unit such as trucks, shovels, and loading or dumping points to act as an autonomous agent capable of adapting to environmental changes and interacting with other agents [17]. This modeling approach is particularly well-suited for capturing the decentralized nature of real mining operations, where individual truck operators make route and speed decisions based on local information, and dispatching systems coordinate fleet movements through distributed communication mechanisms.

Prior research in mining logistics shows that decentralized coordination mechanisms can significantly improve fleet synchronization and reduce idle time [18], [19]. Studies comparing centralized versus decentralized dispatching strategies have demonstrated that agent-based approaches can achieve near-optimal performance while offering greater flexibility and robustness to disruptions [20]. Furthermore, ABM frameworks can naturally incorporate learning mechanisms, allowing agents to improve their decision-making over time based on accumulated experience [21]. Recent advances in multi-agent reinforcement learning have shown particular promise for real-time truck dispatching, where adaptive algorithms dynamically adjust routing decisions based on current system states and predicted future conditions [20].

Further evidence from transportation, logistics, and supply chain systems demonstrates ABM's suitability for modeling environments characterized by heterogeneity, stochasticity, and dynamic interactions [22], [23]. Applications of ABM in urban transportation have shown its effectiveness in capturing traffic flow dynamics and route choice behavior. In logistics systems, agent-based approaches have been used to model warehouse operations, distribution networks, and horizontal cooperation among carriers. Supply chain applications have leveraged ABM to analyze resilience mechanisms and evaluate risk mitigation strategies under uncertainty.

Within the mining domain, ABM has been used to evaluate multimodal transportation systems, dynamic dispatching strategies, and bottleneck behavior under uncertainty [24]. However, despite its advantages, the application of ABM in Iranian open-pit mines remains limited. Many domestic studies still rely on deterministic or DES-based frameworks that omit key features such as variability in equipment availability, heterogeneous fleet characteristics, and multi-layer operational constraints [5]. Consequently, these models often fail to accurately capture the true complexity of real haulage systems, particularly in large-scale mines such as Golgohar.

In this study, an Agent-Based Simulation (ABS) model is developed for Golgohar Iron Ore Mine No. 1 to represent real-time interactions among trucks, shovels, excavators, and dumping points under stochastic operational conditions. The model incorporates actual mine data including availability coefficients, failure–repair patterns, haul distances, and cycle-time variations and enables simulation-optimization evaluation of fleet configurations and availability improvement strategies.

Based on the reviewed literature, the key research gaps can be summarized as follows:

- Existing haulage optimization frameworks rarely integrate decentralized decision-making, adaptive behavior, and system-level interactions within a unified and scalable modeling environment.
- Many agent-based studies oversimplify stochastic behaviors such as breakdown events, queue evolution, and dynamic rerouting, reducing their ability to represent real operational complexity.
- Most prior models rely on fixed or idealized equipment availability assumptions and fail to incorporate heterogeneous and variable availability patterns observed in real fleets.
- Critical operational constraints including route accessibility limits, shovel capacity variability, and heterogeneous truck types are seldom modeled simultaneously within a single integrated framework.
- Traditional scenario-based studies often evaluate haulage parameters independently and lack a systematic optimization mechanism. In contrast, the present simulation–optimization framework simultaneously considers truck availability, shovel availability, stochastic cycle-time variability, and queue constraints, capturing their combined and nonlinear effects on overall system performance and operational balance.

Table1. Literature review of research.

Ref	Year	Journal	Methodology	Core Contribution	Main Gap Filled by Current Study
[1]	2021	Resources Policy	Systematic review (200+ papers)	First comprehensive map of mine supply chain management topics	Lacked decentralized, stochastic agent modeling of upstream haulage
[8]	2020	IEEE Transactions on Engineering Management	Conceptual measurement framework	Supply chain resilience in open systems	Conceptual only; no operational ABS implementation
[29]	2022	Computers & Industrial Engineering	Multi-objective optimization + simulation	Sustainable CLSC network design with inflation & carbon policies	General CLSC; no mining-specific stochastic equipment agents
[30]	2022	Decision	Neutrosophic	Optimal sustainable	Theoretical; no real-

Ref	Year	Journal	Methodology	Core Contribution	Main Gap Filled by Current Study
		Making: Applications in Management and Engineering	programming	CLSC under uncertainty	mine ABS validation
[31]	2023	Journal of Decisions and Operations Research	Heuristic hybrid metaheuristic	Sustainable CLSC with inflation & emissions	Network level only; ignores real-time fleet dynamics
[11]	2024	International Journal of Production Research	Process-mining review	State-of-the-art in supply chain process mining	Calls for agent-based extensions beyond event logs
[10]	2024	Resources Policy	50-year bibliometric analysis	Outlines future priorities: digitalization, resilience, integrated optimization	No high-fidelity ABS validation against real mine data
[20]	2024	Computers & Operations Research	Deep RL + simulation	Real-time adaptive truck dispatching	Centralized RL; no fully decentralized stochastic availability
Current Study	2026	(Resources Policy / Computers & Operations Research)	Agent-Based Simulation + NSGA-II simulation-optimization + Monte Carlo	First data-driven ABS of Golgohar's supply chain haulage using real heterogeneous availability, decentralized agents, queue dynamics, and joint truck–shovel optimization	Closes all listed gaps: validated (±2.5 % error), context-specific, reveals truck bottleneck, and proves 30 %/10 % coordinated improvement yields +31.4 % extraction with –13 % queue

These gaps highlight the need for a high-fidelity, data-driven simulation framework capable of capturing the interactive, uncertain, and decentralized nature of haulage operations in complex open-pit mines. In addition, in contrast to conventional approaches, discrete-event simulation (DES) models the system as a sequence of predefined events and lacks autonomous, decision-making entities capable of local adaptation. Stochastic optimization methods typically operate in an ex-ante manner, where decisions are optimized prior to execution based on assumed probability distributions and remains largely non-adaptive during runtime. In comparison, the agent-based modeling framework adopted in this study represents trucks and shovels as decentralized, autonomous agents that dynamically adapt their decisions in response to stochastic operational conditions and spatial system states, thereby enabling the emergence of realistic, system-level behavior.

3. Problem statement. This study provides several key contributions to the literature on open-pit haulage optimization and agent-based mining simulation:

- Context-specific innovation for heterogeneous Iranian open-pit operations: To the authors' knowledge, this study is the first to implement a data-driven Agent-Based Simulation of truck–shovel operations calibrated with real operational data from an Iranian open-pit mine. Unlike previous Iranian studies that relied on deterministic or centralized models, the proposed framework explicitly models heterogeneous fleets, contractor-dependent availability coefficients, and

stochastic behavior, reflecting the complex and variable conditions of Iranian mining operations. This integration of real reliability data and decentralized decision logic represents both a methodological and contextual novelty within the region's mining research.

- Development of a unified, decentralized agent-based haulage framework: Unlike traditional discrete-event or centralized dispatching models, this research constructs a fully decentralized Agent-Based Simulation (ABS) architecture in which trucks, shovels, loaders, and dumping points act as autonomous agents governed by adaptive behavioral rules. This enables the model to capture emergent fleet dynamics and localized decision-making, a feature absent in earlier works.

- Data-driven integration of stochastic operational realities: Unlike many previous studies that relied on idealized or fixed availability assumptions, the present research integrates real operational data from Golgohar Mine No. 1, including equipment availability coefficients, failure and repair dynamics, and cycle time variability. This data driven approach enables the simulation model to capture the inherent stochasticity and mechanical heterogeneity of the actual mining fleet with higher fidelity. Although related studies such as Hazrathosseini and have explored reliability enhancement using reinforcement learning, the systematic incorporation of real operational reliability data within an agent based simulation framework has received limited attention in the existing literature.

- Simulation–optimization under joint subsystem variability: While previous agent-based or stochastic studies typically assessed truck or shovel improvements independently the present research integrates a validated agent-based simulation model with a multi-objective simulation–optimization framework. This approach systematically evaluates the combined impact of truck and shovel availability improvements under stochastic operational conditions and logistical constraints. By explicitly incorporating queue limitations and real reliability data, the model captures the nonlinear interdependencies between the loading and hauling subsystems and identifies a coordinated improvement strategy that achieves maximum extraction while maintaining operational balance.

- Incorporation of spatially and operationally realistic constraints: Through the use of a spatially explicit mine layout and accessibility-based dispatch logic, the model realistically replicates haulage routes, queue evolution, and congestion patterns. This spatial fidelity extends beyond the simplified network assumptions commonly used in optimization studies.

- A validated decision-support and research foundation for intelligent mining: Validation using ten months of actual operational data demonstrates that the proposed agent based simulation model achieves a deviation of less than 2.5 percent from real mine production. This level of accuracy supports the applicability of the model as a practical decision support tool for fleet management while also providing a robust foundation for future research extensions.

3.1. Paper organization. The remainder of this paper is organized as follows. Section 2 introduces the methodology. Section 3 presents the simulation–optimization approach

aimed at increasing ore and waste extraction while satisfying operational and logistical constraints. Section 4 provides the discussion. Finally, Section 5 concludes the paper.

4. Methodology.

4.1. Agent-based simulation. When a system exhibits dynamic behaviors with temporal dependencies and complex interactions among its components, Agent-Based Modeling and Simulation (ABMS) provides a suitable approach for representing and analyzing such processes. In ABMS, the overall system behavior emerges from the interactions of autonomous decision-making entities called agents, each possessing individual attributes, behaviors, and goals. In ABM, an entity qualifies as an agent when it demonstrates four essential characteristics: autonomy, situatedness, interactivity and adaptivity. Autonomy implies that the agent operates independently, making decisions based on internal rules without centralized control. Situatedness means the agent is embedded in a specific environment from which it perceives stimuli and to which it reacts [25]. Interactivity encompasses the agent's ability to communicate and coordinate with other agents or the environment through direct or indirect interactions [26]. Finally, adaptivity reflects the capacity to learn or change behavior in response to environmental dynamics or interactions over time, a critical factor in simulating emergent system behavior [27], [28]. These properties make agents suitable representations for modeling decentralized, dynamic, and complex systems across domains such as mining, economics, and logistics.

4.2. Agent-based simulation for Golgohar Mine. This research leverages the ABM paradigm to examine the operational dynamics of iron ore extraction at Golgohar Mine No. 1, owing to ABM's adaptability and ability to capture nuanced behaviors of system components. An agent-based simulation model was constructed using the Any Logic software, facilitating the integration of dynamic, discrete-event, and agent-based modeling methodologies.

A comprehensive examination of existing literature on agent attributes and a meticulous analysis of mining processes and stakeholders informed the identification of key agents for model inclusion, specifically:

- Mine Agent: Representing the mine's overall operational context
- Truck Agent: Ore and waste transportation processes
- Shovel Agent: Capturing the loading activities
- Dispatcher Agent: Coordination and allocation of resources.

4.2.1. Truck agent. The behavior of the Truck Agent is implemented using statecharts within the AnyLogic simulation environment. Initially, the truck remains in a "Ready" state, waiting to receive an assignment. Upon receiving a message from the Dispatcher Agent indicating a designated working face, the truck transitions to the "Move To" state and navigates toward the assigned shovel location. Once it arrives at the working face, if the corresponding Shovel Agent is idle, the "Loading" process is initiated and the truck's state changes accordingly. If the shovel is currently busy, the truck waits until the shovel becomes available. Upon completion of the loading process, the shovel sends a message to the truck indicating that loading is finished. Depending on the type of material loaded (iron ore or waste rock), the truck transitions to the "Move to Dest" state and proceeds to the corresponding destination, either the crusher or the waste dump. After arrival, the truck enters the "Unloading" state, where the material is discharged. Upon completing the

unloading operation, the truck returns to the “Ready” state, awaiting the next assignment from the dispatcher.

Additionally, the Truck Agent is subject to probabilistic failure events. If a breakdown occurs, the truck transitions to the “Breakdown” state and remain inactive for a repair duration specified by system parameters. After repairs are completed, the truck returns to the “Ready” state, resuming its operational cycle. The developed state chart diagram, based on the described behavior, is presented in the figure (1) below within the any logic simulation environment.

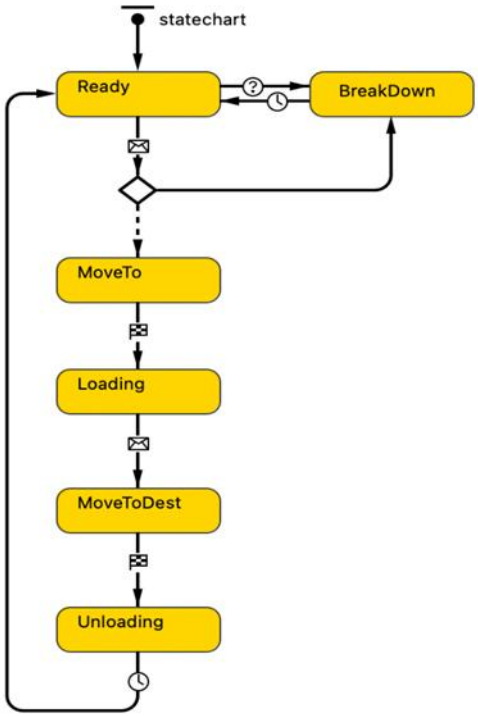


Figure 1. State-transition chart of the Truck.

4.2.2. Shovel agent. The Shovel Agent is responsible for loading material into the haul trucks. Similar to the Truck Agent, its behavior is implemented using state charts in the any logic simulation environment. Initially, the shovel is in a “Ready” state, indicating it is available for assignment. Once it is allocated to a working face, its state transitions to “Wait for Working”, indicating that it is positioned and ready to begin loading operations.

If the Truck agent arrives at the shovel’s location, the shovel changes its state to “Start Loading”, initiating the loading process. The duration of loading is dependent on the truck’s capacity and is dynamically retrieved from the truck. Upon completion, the shovel sends a message to the truck indicating the end of the loading operation and transitions to the “Finish Loading” state. If another truck is queued and waiting to be loaded, the shovel immediately resumes loading. otherwise, it remains idle in “Wait for Working” until the arrival of the next truck. This state-driven approach enables the dynamic coordination between shovels and trucks under various operational conditions.

The implemented state chart reflecting this behavior within the Any Logic platform is illustrated in the figure (2) below.

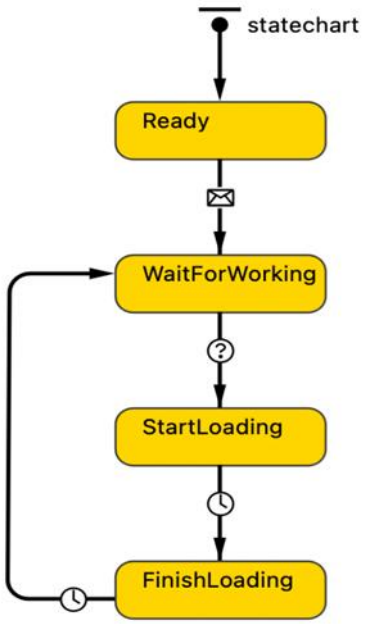


Figure 2. State-transition chart of the Shovel.

4.2.3. Dispatcher agent. The dispatcher agent is responsible for two primary functions: the initial allocation of shovels to working faces at the start of the simulation, and the dynamic assignment of haul trucks to loading locations during simulation runtime. Various dispatching policies can be implemented within this agent, and in this study, queue length has been adopted as a key decision-making criterion. When a truck agent enters the ready state, the Dispatcher identifies all currently active working faces that are compatible with the truck type (e.g., ore or waste). It then monitors the level of congestion and queue length at each potential loading site. Based on this information, the dispatcher selects the most suitable working face, typically the one with the shortest queue and distance, and sends a message to the Truck agent, instructing it to proceed to the assigned location. This mechanism ensures a balanced distribution of haulage tasks and minimizes waiting times, thereby improving overall system efficiency.

4.2.4. Mine agent. The mine agent functions as the environmental framework within which all other agents, including trucks, and shovels operate. In the simulation model, the mine environment encompasses key spatial and operational components such as haulage routes, working faces, waste dump sites, crusher locations, parking areas, and maintenance workshops. All these components were implemented within the Any Logic simulation environment using a realistic spatial layout (layout-based modeling), ensuring that path lengths and positions reflect the actual geometry of Golgohar Iron Ore Mine No. 1. The working faces are predefined zones where shovel agents are positioned, and truck agents navigate toward these locations based on assignments received from the dispatcher agent. Route selection and travel time are determined by available paths and truck-

specific speed profiles, enabling accurate modeling of movement dynamics and spatial constraints.

A visual representation of the Mine Agent within the any logic simulation environment is shown in the figure (3) below:

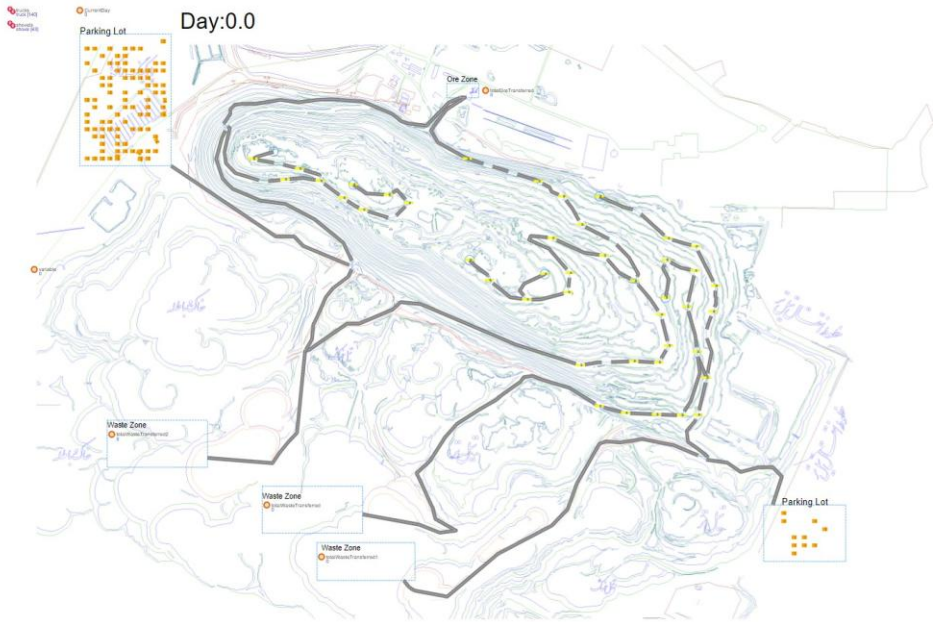


Figure 3. A view of the Golgohar Mine No. 1 simulation model in the AnyLogic environment.

4.2.5. Input parameters. Several key parameters were considered as input variables in the simulation model:

- Number, type, and capacity of trucks
- Number and capacity of shovels
- Truck travel speed, categorized by truck type and capacity
- Loading time, specified by truck type and capacity
- Downtime and repair durations
- Working faces
- Unloading time, categorized by truck type

The agent-based simulation was executed for a defined operational period, during which the state of each agent was iteratively updated at each simulation step.

5. Numerical study.

5.1 Input data. The agent-based simulation model was parameterized using a comprehensive set of operational inputs obtained from Golgohar Iron Ore Mine No. 1. These inputs include the number and capacity of haul trucks, the specifications of loading equipment (shovels, excavators, and loaders), and their corresponding availability coefficients. All data were sourced from the mine's operational database and performance

records. The availability coefficient for each equipment type represents the ratio of its actual operating time to the total calendar time during the production period. This metric reflects the real-world reliability and readiness of equipment under normal mine conditions.

5.1.1. Truck fleet characteristics. The haulage system at Golgohar Mine No. 1 is operated by two contracting companies, each responsible for a distinct portion of the truck fleet. Tables (2) and (3) present the capacity, quantity, and availability characteristics of the fleets managed by each contractor. The substantial variation in availability coefficients across truck types highlights differences in equipment age, mechanical condition, and maintenance performance within the fleet.

Table 2. Contractor Company 1 Trucks Type.

Truck Type	*Quantity*	*Capacity (ton)*	*Availability Coefficient*
Type 1	11	100	0.70
Type 2	17	100	0.41
Type 3	15	100	0.66
Type 4	20	40	0.21
Type 5	36	100	0.48
Type 6	33	55	0.33
Type 7	14	60	0.12

Table 3. Contractor Company 2 Trucks Type.

Truck Type	*Quantity*	*Capacity (ton)*	*Availability Coefficient*
Type 1	11	100	0.70
Type 2	17	100	0.41
Type 3	15	100	0.66
Type 4	20	40	0.21
Type 5	36	100	0.48
Type 6	33	55	0.33
Type 7	14	60	0.12

5.1.2. Loading equipment characteristics. The loading system comprises shovels, excavators, and loaders operated by both contractors. The specifications and availability coefficients of these units are provided in Tables (4) and (5). The diversity in equipment capacity and availability influences loading cycle duration, material handling efficiency, and queue formation at active faces.

Table 4. Contractor Company 1 Shovels Type.

Equipment Type	*Quantity*	*Capacity (m^3)*	*Availability Coefficient*
Shovel Type 1	10	4.0	0.24
Shovel Type 2	6	4.0	0.57
Shovel Type 3	2	3.6	0.33
Shovel Type 4	2	3.6	0.56
Shovel Type 5	1	3.6	0.53
Shovel Type 6	9	4.0	0.61

Equipment Type	*Quantity*	*Capacity (m³)*	*Availability Coefficient*
Shovel Type 7	1	5.0	0.09
Shovel Type 8	7	5.0	0.38
Shovel Type 9	5	4.5	0.17
Shovel Type 10	2	4.2	0.49

Table 5. Contractor Company 2 Shovels Type.

Equipment Type	*Quantity*	*Capacity (m³)*	*Availability Coefficient*
Shovel Type 1	2	4.5	0.59
Shovel Type 2	4	10.0	0.56
Shovel Type 3	5	8.0	0.56
Shovel Type 4	2	12.0	0.15
Shovel Type 5	1	4.5	0.80

These datasets form the foundation for defining agent attributes and model input parameters. Given the stochastic nature of equipment failures and maintenance events, availability coefficients are incorporated as key reliability indicators affecting both truck–shovel assignment and overall system performance.

5.1.3. Occupancy ratios and conversion factors. To account for voids, present during bucket loading and material transport, specific occupancy ratios were applied in the model:

- Truck bed occupancy: 80%
- Shovel, excavator, and loader bucket occupancy: 85%

Bucket capacities were converted to material weight using density-based conversion factors representative of site conditions:

- Iron ore: 4.2 t/m³
- Waste: 2.35 t/m³

These conversion factors reflect the in-situ density of ore and waste at Golgohar and ensure realistic mass–volume translation in the simulation.

5.1.4. Mine layout and route configuration. The spatial layout of Golgohar Mine No. 1 including haulage roads, working faces, loading points, and dumping areas—was fully integrated into the simulation environment. All distance and speed parameters were calibrated to match real operating conditions, enabling accurate replication of truck cycle times, queue dynamics, and fleet movement patterns. This integration supports precise evaluation of system delays, route-related variability, and loading–hauling coordination.

5.1.5. Truck allocation policy. A dynamic dispatching policy was implemented to assign trucks to loading points. The allocation decision was based on:

i. The current queue length at each loading unit, and
ii. Route accessibility constraints for light and heavy trucks.

When a truck completed a haul cycle, it was automatically assigned to the working face with the shortest queue and with authorized access. This adaptive policy minimized truck idle time and contributed to higher overall fleet efficiency within the simulated mine environment.

5.1.6. Model validation. To validate the simulation model, actual production data for iron ore and waste haulage were collected for a 10-month period (April 2023–January 2024) and compared with the simulated output for the same timeframe. Table (6) summarizes the comparison between actual and simulated extraction levels.

Table 6. Actual value vs. simulated results of iron ore and waste extracted.

Parameter	*Actual Value (tons)*	*Simulated Value (tons)*
Iron ore extracted (10-month period)	8,688,293	8,630,000
Waste material extracted (10-month period)	70,615,962	68,870,000

The results show a deviation of less than 2.5% between simulated and real production values, demonstrating the high accuracy and reliability of the developed agent-based model Figure (4). This strong alignment confirms that the model effectively captures haulage dynamics, equipment performance variability, and operational interactions, thereby providing a robust analytical framework for fleet allocation, scheduling, and strategic decision-making in Golgohar Mine No. 1.

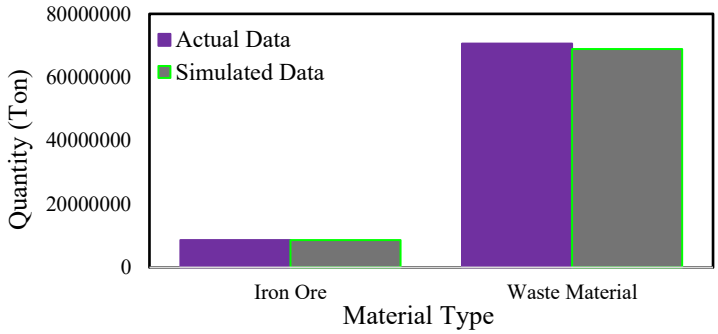


Figure 4. Comparison between actual and simulated extraction volumes for iron ore and waste material in Golgohar Mine No. 1.

Based on the successful validation of the simulation model, the remaining outputs of the model under the current (baseline) condition are presented in the following section.

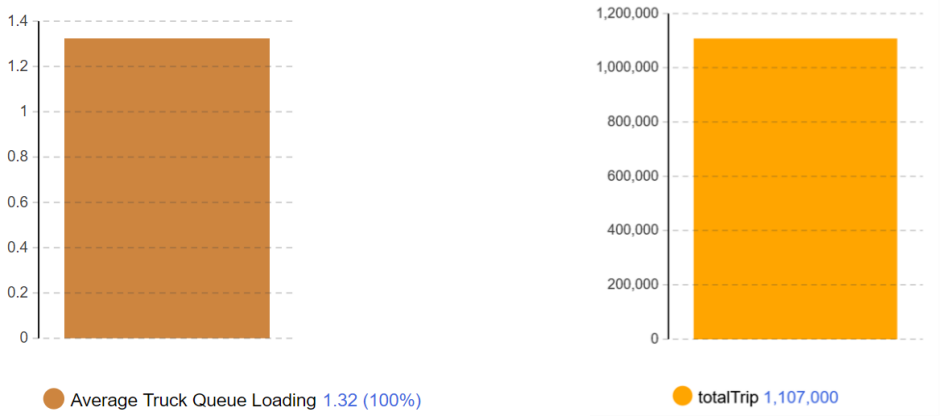


Figure 5. The simulation model outputs.

As illustrated in the figures (5), the operational performance of the haulage system under the current (baseline) condition can be clearly observed from the validated agent-based simulation model outputs. The results show that the average truck queue length at loading points is approximately 1.32 trucks. Although this level of congestion is not critical, it reflects the presence of waiting times that may reduce overall operational efficiency. In addition, the total number of truck trips during the simulation horizon is approximately 1,107,000 cycles. This value represents the overall throughput of the haulage system and is directly influenced by equipment availability and coordination between trucks and shovels.

5.2. Simulation based optimization. In this study, the validated agent-based simulation model developed in AnyLogic is integrated with an optimization procedure to form a simulation–optimization framework aimed at increasing ore and waste extraction while considering operational and logistical constraints. The core idea is to treat the simulation model as a high-fidelity stochastic evaluator, while the optimization algorithm systematically explores the decision space to identify optimal or near-optimal operating policies.

After model verification and validation, the simulation model is provided as an evaluation engine to the optimization module in AnyLogic. The Non-dominated Sorting Genetic Algorithm II (NSGA-II) is employed as the search mechanism due to its robustness in handling nonlinear, stochastic, and multi-objective problems. In each generation, NSGA-II generates a population of candidate solutions, where each individual corresponds to a specific combination of decision variable values (e.g., improvement levels in truck and shovel availability).

For each candidate solution, the following procedure is executed:

i. The decision variables are assigned to the corresponding input parameters of the simulation model.
ii. The simulation model is executed over the defined planning horizon.
iii. The model returns the objective function values, such as total ore production, total waste removal, equipment utilization, average queue length, or other relevant performance indicators.
iv. These objective values are passed back to the NSGA-II algorithm for fitness evaluation and non-dominated sorting.

In line with the above simulation–optimization interaction, the problem is formally defined as a multi-objective mathematical programming model. The decision variables, objective functions, and constraints are defined as follows.

$$Max\ Z = \sum_{t=1}^{T} (P_{ore,t} + P_{waste,t}) \tag{1}$$

Subject to:

$$0 \le \delta_T \le \delta_T^{max} \tag{2}$$

$$0 \le \delta_S \le \delta_S^{max} \tag{3}$$

$$\frac{1}{T}\sum_{t=1}^{T} Q_t \le Q_{max} \tag{4}$$

where:

T number of time periods in the simulation horizon (5)

$P_{ore,t}$ amount of extracted ore in period t (6)

$P_{waste,t}$ amount of waste removed in period t (7)

δ_T^{max}	maximum allowable Truck availability improvement	(8)
δ_S^{max}	maximum allowable Shovel availability improvement	(9)
Q_t	average truck queue length at loading points in period t	(10)
Q_{max}	maximum allowable average queue length	(11)

The decision variables of the model are the availability improvement factors of trucks and shovels, defined as follows:

δ_T	percentage increase in truck availability	(12)
δ_S	percentage increase in shovel availability	(13)
$P_{ore,t}$, $P_{waste,t}$, Q_t	are obtained from the validated AnyLogic simulation model under the decision vector (δ_T, δ_S).	(14)

The above mathematical model, including the objective function, constraints, and decision variables, was implemented in the AnyLogic simulation environment. The validated simulation model was integrated with the optimization experiment module, where the decision variables δ_T and δ_S were defined as controllable input parameters within their specified bounds. The objective function and constraint (queue length limit) were evaluated based on the simulation outputs for each candidate solution generated by the optimization algorithm.

For the optimization process, a total of 5000 iterations were considered. This number was selected based on convergence behavior: beyond 5000 iterations, no significant improvement in the objective function value or the Pareto front quality was observed. Therefore, 5000 iterations were deemed sufficient to ensure solution stability and near-optimal performance while avoiding unnecessary computational effort. To ensure the reproducibility and computational reliability of the results, the proposed Agent-Based Simulation (ABS) model was developed and executed on a high-performance workstation equipped with an Intel Core i7-12700K processor (3.61 GHz), 64 GB of DDR4 RAM. The model was implemented using AnyLogic Professional Edition 8.9.1, running on a 64-bit Windows 11 Pro operating system.

In the optimization model, the maximum achievable improvement in truck availability was assumed to be 30%. Accordingly, the upper bound of the truck availability improvement factor was set to $\delta_T^{max} = 0.30$.

In addition, the maximum allowable average truck queue length at loading points was limited to 2 units. Therefore, the queue constraint was defined such that the mean truck queue length during the simulation horizon does not exceed $Q_{\max} = 1.5$.

The output of the optimization process indicated that the optimal values of the decision variables were obtained as follows:

$$\delta_T^* = 0.3, \qquad \delta_S^* = 0.1.$$

To determine the improvement in system performance under the optimized decision variables, the updated availability values were incorporated into the validated agent-based simulation model. Subsequently, a Monte Carlo simulation was performed, in which the

model was executed for 1000 replications using different random number seeds. The average value of the objective function was then calculated across all replications to ensure statistical reliability of the results.

The outputs of the updated simulation model, together with the results of the simulation–optimization process, are presented in the figures (6) below.

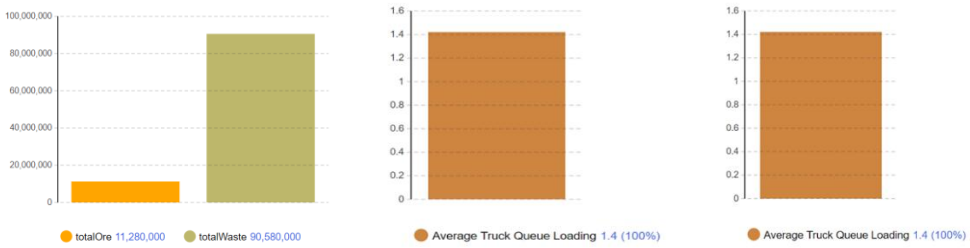


Figure 6. The simulation model outputs related to fleet utilization.

As shown in the figures, under the optimized configuration (δ_T^* = 0.30 and δ_S^*= 0.10), total ore production reaches 11,280,000 tons, while total waste removal increases to 90,580,000 tons. This reflects a significant improvement in overall extraction performance compared to the baseline condition. The average truck queue length at loading points is reduced to 1.4 trucks, indicating improved coordination between the hauling and loading subsystems. Despite the higher production level and increased number of haulage cycles, congestion remains controlled and below the defined operational constraint. To enable a clearer comparison between the results of the simulation model under the current (baseline) condition and those obtained from the simulation–optimization process; a comparative summary is presented in the table below.

Table 7. The result comparison of baseline and optimized simulation model.

Indicator	Baseline	Optimized	Improvement (Δ)	Improvement (%)
δ_T	0.00	0.30	+0.30	—
δ_S	0.00	0.10	+0.10	—
P_{ore} (Mt)	8.63	11.28	+2.65	+30.7%
P_{waste} (Mt)	68.87	90.58	+21.71	+31.5%
Z = P_{ore}+ P_{waste} (Mt)	77.50	101.86	+24.36	+31.4%
Avg. Truck Queue	1.61	1.40	−0.21	−13.0%
Total Trips (million)	1.10	1.465	+0.365	+33.2%

The results presented in Table (7) clearly demonstrate the effectiveness of the proposed simulation–optimization framework.

Under the baseline condition, where no improvement in equipment availability is applied (δ_T = 0.00 and δ_S = 0.00), the total extracted material amounts to 77.50 Mt, consisting of 8.63 Mt of ore and 68.87 Mt of waste. The average truck queue length in this state is 1.61.

After applying the optimized decision variables (δ_T = 0.30 and δ_S = 0.10), a substantial improvement in system performance is observed. Ore production increases by 2.65 Mt (approximately 30.7%), while waste removal increases by 21.71 Mt (approximately 31.5%). Consequently, the total extracted material (Z) rises from 77.50 Mt to 101.86 Mt, representing an overall improvement of about 31.4%.

Importantly, this increase in production is achieved while simultaneously reducing congestion in the system. The average truck queue length decreases from 1.61 to 1.40, corresponding to a 13% reduction. This indicates that the optimized configuration not only enhances throughput but also improves operational balance between the hauling and loading subsystems.

Overall, the results confirm that coordinated improvement of truck and shovel availability leads to a more efficient and balanced haulage system, eliminating the bottleneck observed in the baseline scenario and significantly increasing total extraction performance without violating the queue constraint.

To further clarify the magnitude of improvement achieved through the optimization process, a sensitivity analysis related to the increase in truck availability is presented in Table (8).

Table 8. Sensitivity analysis of truck availability improvement in the simulation model.

Performance Indicator	**Baseline**	**Truck + 5%**	**Truck + 10%**	**Truck + 15%**	**Truck + 20%**	**Truck + 30%**	**Combined (Truck + 30% / Shovel 10%)**
P_{ore} (Mt)	8.63	9.15	9.342	9.836	10.26	10.94	11.28
P_{waste} (Mt)	68.87	73.19	75.37	79.02	81.93	87.54	90.58
Average Truck Queue	1.61	1.42	1.57	1.57	1.61	1.82	1.40
Total Trips (million)	1.107	1.170	1.218	1.271	1.313	1.401	1.645

As shown in Table (8), increasing truck availability alone leads to higher ore and waste production; however, beyond a certain level of improvement, system congestion begins to intensify. In particular, when truck availability is increased to 30% without simultaneous improvement in shovel availability, the average truck queue rises to 1.82, exceeding the desirable operational level and indicating the emergence of a loading bottleneck.

Although extraction continues to increase with higher truck availability, the growing queue length reflects reduced system balance and increasing congestion at loading points. This demonstrates that improving truck availability alone does not guarantee efficient system performance and may shift the bottleneck toward the shovel subsystem.

Therefore, the results confirm that the simulation–optimization process has successfully achieved operational balance while maximizing total extraction and simultaneously satisfying logistical and operational constraints.

6. Discussion. In this study, an Agent-Based Simulation (ABS) model was developed in the Any Logic environment to analyze and optimize the extraction operations of Golgohar Iron Ore Mine No. 1. By modeling dynamic interactions among trucks, shovels, excavators, loading points, and working faces and by integrating a queue-based dispatching policy with haul-road accessibility constraints the model successfully captured the realistic operational behavior of the mine. Validation using ten months of actual production data (April 2023–January 2024) demonstrated less than 2.5% deviation, confirming the model's accuracy, reliability, and capacity to replicate real-world mining dynamics.

6.1. Key findings. Analysis of the simulation–optimization results and the associated sensitivity evaluations yielded several important insights regarding system bottlenecks, capacity balance, and operational stability:

- The primary system bottleneck is located within the truck fleet under baseline conditions. Increasing shovel availability alone does not improve production, since the hauling subsystem constrains overall throughput. This confirms that the system is haulage-limited rather than loading-limited in its initial configuration.
- The optimal configuration identified through the simulation–optimization framework corresponds to a 30% increase in truck availability combined with a 10% increase in shovel availability. This coordinated improvement results in the highest total extraction levels (11.28 Mt of iron ore and 90.58 Mt of waste) while simultaneously reducing the average truck queue length at loading points. Unlike single-parameter improvements, this solution enhances throughput without creating additional congestion.
- Sensitivity analysis reveals that further increasing truck availability without proportional adjustment of shovel capacity leads to rising queue lengths and congestion. Similarly, excessive enhancement of shovel availability yields diminishing returns. These findings demonstrate that unbalanced capacity expansion does not eliminate bottlenecks but merely shifts them across subsystems, reducing operational stability.
- Operational indicators including utilization rates, truck queue lengths, and total haulage cycles showed behaviors consistent with engineering logic and real site constraints. This confirms the internal coherence of the decision-making architecture embedded in the simulation model.

6.2. Managerial recommendations. Based on the optimization outcomes, the following managerial actions are recommended to improve system efficiency and operational stability:

- Prioritize improvements in truck availability through predictive and condition-based maintenance programs, critical spare-parts management, optimized service scheduling, and mitigation of recurrent failure modes.
- Implement a moderate increase (approximately 10%) in shovel availability to maintain balance with enhanced hauling capacity and avoid unnecessary overinvestment in loading equipment.
- Refine truck dispatching strategies by incorporating queue-length monitoring, haul-road restrictions, and near real-time updates to minimize mutual idle time between trucks and shovels.
- Utilize the ABS model as a decision-support tool to rapidly test fleet configurations, investment scenarios, and operational policies before real-world implementation.

7. Conclusion. This study developed and validated a data-driven agent-based simulation framework to analyze and optimize truck–shovel haulage operations in a large-scale open-pit mining environment. By explicitly modeling trucks, shovels, dispatching logic, and spatial mine layout as interacting autonomous agents, the proposed framework captures the decentralized, stochastic, and dynamic nature of real-world haulage systems more realistically than conventional deterministic or discrete-event approaches.

The simulation model was calibrated using ten months of actual operational data from Golgohar Iron Ore Mine No. 1, including heterogeneous fleet characteristics, equipment availability coefficients, failure–repair dynamics, and route-specific constraints.

Validation results demonstrated a deviation of less than 2.5% between simulated and actual production levels for both iron ore and waste material, confirming the robustness and practical reliability of the proposed agent-based modeling approach.

Based on the results obtained from the simulation–optimization framework and subsequent Monte Carlo evaluation, several important insights were identified regarding system behavior and bottleneck dynamics. The findings confirm that the primary performance bottleneck of the studied haulage system lies within the hauling subsystem under baseline conditions. Incremental improvements in truck availability led to noticeable increases in ore and waste production; however, when truck availability was enhanced in isolation, queue lengths at loading points increased, indicating a transfer of the bottleneck toward the loading subsystem.

In contrast, isolated improvements in shovel availability produced limited impact on overall system throughput under the existing hauling constraints, confirming that the system was initially haulage-limited. The most substantial and operationally stable performance improvements were achieved through the coordinated enhancement strategy identified by the simulation–optimization process. Specifically, a 30% increase in truck availability combined with a 10% increase in shovel availability resulted in a 31.4% increase in total extraction, a 33.2% increase in total haulage cycles, and a 13% reduction in average truck queue length.

These findings highlight a critical managerial insight. Single-subsystem optimization strategies can shift system bottlenecks without resolving underlying inefficiencies, whereas integrated availability management across interacting subsystems yields sustainable performance gains. From an operational perspective, the results support prioritizing truck availability improvements through reliability-centered maintenance, predictive diagnostics, and optimized service planning, while simultaneously maintaining shovel availability at levels sufficient to absorb increased hauling capacity.

Despite its strengths, the present study is subject to several limitations. The dispatching mechanism is based on a rule-based, myopic decision logic that relies on current system states and does not incorporate predictive or learning-based behavior. Equipment reliability is represented through aggregated availability coefficients and stochastic downtime distributions, without explicitly modeling detailed failure modes or condition-based degradation. Additionally, human factors such as operator behavior variability and organizational constraints are not explicitly represented, and economic or environmental performance indicators, such as fuel consumption, emissions, or maintenance costs, are not directly quantified.

These limitations provide clear directions for future research. Subsequent studies may extend the proposed framework by integrating learning-enabled dispatching strategies, such as multi-agent reinforcement learning, to enhance anticipatory decision-making. Incorporating more detailed reliability and maintenance models, including condition-monitoring data, would further improve diagnostic accuracy and operational realism. Future work may also expand the model to include economic and sustainability metrics, enabling multi-objective optimization of productivity, cost, and environmental performance. Finally, applying the framework to multiple mining sites with varying geological and operational characteristics would support broader generalization and comparative analysis across different open-pit mining contexts.

- **i.** The ABS model, built in Any Logic and calibrated with ten months of real Golgohar data (April 2023–January 2024), matched actual production within 2.5 %—iron ore 8.63 Mt simulated vs. 8.69 Mt real; waste 68.87 Mt vs. 70.62 Mt—proving it faithfully reflects on-site reality.
- **ii.** In the baseline (no improvements), the supply chain moved 77.50 Mt total material with 1.107 million truck trips and an average queue of 1.61 trucks; the clear bottleneck was truck availability.

iii. The NSGA-II optimization pinpointed the sweet spot: +30 % truck availability and +10 % shovel availability. This lifted ore to 11.28 Mt (+30.7 %), waste to 90.58 Mt (+31.5 %), and total material to 101.86 Mt (+31.4 %), while dropping the average queue to 1.40 trucks (–13 %).

iv. Improving trucks alone beyond 20 % simply moved the bottleneck to shovels and pushed queues above 1.8; improving shovels in isolation barely helped—only the balanced 30/10 combination delivered both higher output and better flow.

v. Sensitivity tables and 1,000-run Monte Carlo validation confirmed the coordinated strategy is robust, statistically significant, and the most operationally stable option for the entire supply chain.

Overall, this study demonstrates that agent-based simulation, when integrated with simulation–optimization techniques and grounded in real operational data and validated against actual production outcomes, provides a powerful and flexible decision-support tool for understanding, diagnosing, and improving complex haulage systems in modern open-pit mining operations.

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